

Raonament Basat en Casos.

Temes avançats de CBR

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Temporal CBR (TCBR)

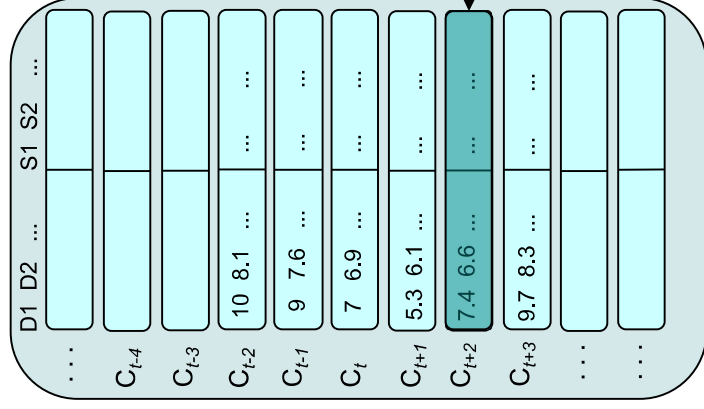
Temporal CBR (TCBR)

The problem (1)

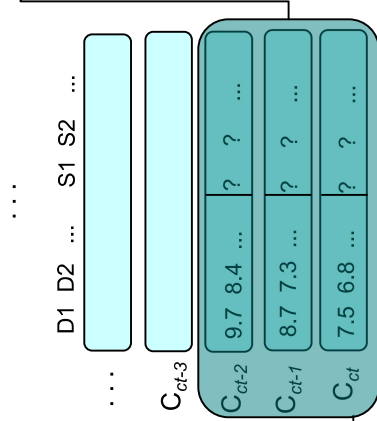
- **Continuous or dynamic or temporal domains**
 - A large amount of new valuable experiences are continuously generated
 - The current state or situation of the domain *depends on previous temporal states or situations* of the domain
 - States have *multiple diagnoses*
- **Examples**
 - Monitoring and on-line control of power stations
 - Wastewater treatment plants control
 - Jet plane control
 - Monitoring of patients in an intensive care unit
 - Diagnosis and/or prognosis and cure of some medical diseases
 - Forecasting of some meteorological or seismic phenomena
 - Autonomous robot navigation control

The problem (2)

Case Library



Current Problems



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EBR Approach

[Sánchez-Marrè *et al.*, 2005] [Pascual-Pañach *et al.*, 2024]

- **Episode-Based Reasoning** is based on the abstraction of temporal sequences of cases, which are named as **episodes**.
- **Open questions**:
 - How to determine the length of an episode?
 - How to represent the episodes, taking into account that they could be overlapping?
 - How to represent the isolated cases?
 - How to relate them to form episodes?
 - How to undertake the episode retrieval?
 - How to evaluate the similarity between temporal episodes of cases?
 - How to continually learn and solve new episodes?

Basic Terminology for EBR (1)

[Sánchez-Marrè *et al.*, 2005] [Pascual-Pañach *et al.*, 2024]

- **Definition 1.** An *isolated case*, or simply *a case* describing several features of a temporal domain at a given moment t , is:

(:case-identifier	CI
:temporal-identifier	t
:case-situation-description	CD
:case-diagnostics-list	CDL
:case-solution-plan	CS
:case-solution-evaluation	CE
)	

Where $CD = ((A_1 \ V_1) (A_2 \ V_2) \dots (A_N \ V_N))$

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Basic Terminology for EBR (2)

[Sánchez-Marrè *et al.*, 2005] [Pascual-Pañach *et al.*, 2024]

- For instance, an *isolated case example* in the domain of volcanic and seismic prediction domain, could be as follows:

(	:case-identifier	CASE-134
:	temporal-identifier	27/11/2004
:	case-situation-description	((SEISMIC-ACT Invaluable) (DEFORMATIONS mean-value) (GEOCHEMICAL-EVOL normal) (ELECT-PHEN level-1))
:	case-diagnostics-list	(No-eruption, Seismic-pre-Alert)
:	case-solution-plan	(Alert-Emergency-Services)
:	case-solution-evaluation	correct
)		

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Basic Terminology for EBR (3)

[Sánchez-Marrè *et al.*, 2005] [Pascual-Pañach *et al.*, 2024]

- **Definition 2.** A *temporal episode* of cases of length l , which is a sequence of l consecutive cases in time, is a structure formed by the following components:

(EI :episode-identifier
 t :initial-time
 l :episode-length
 ED :episode-description
 d :episode-diagnosis
 ES :episode-solution-plan
 EE :episode-solution-evaluation
 C_t :initial-case
 C_{t+l-1} :final-case
)

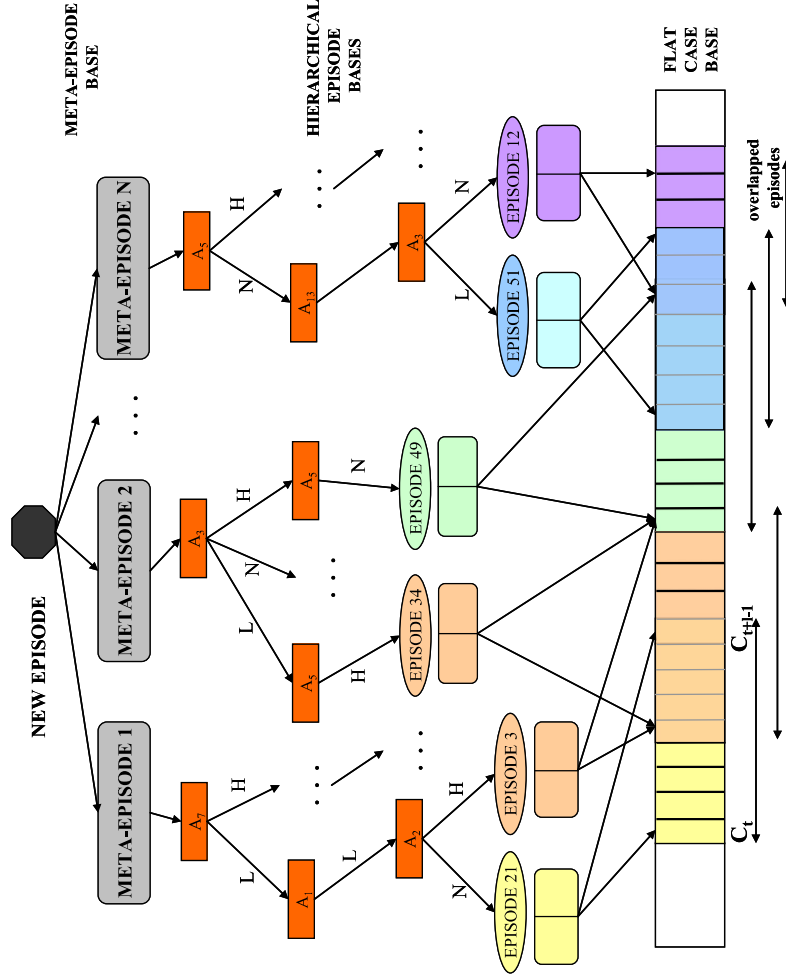
$$E_{t,l}^d = [C_t, C_{t+1}, C_{t+2}, \dots, C_{t+l-1}]$$

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EBR Memory Model

[Sánchez-Marrè *et al.*, 2005]



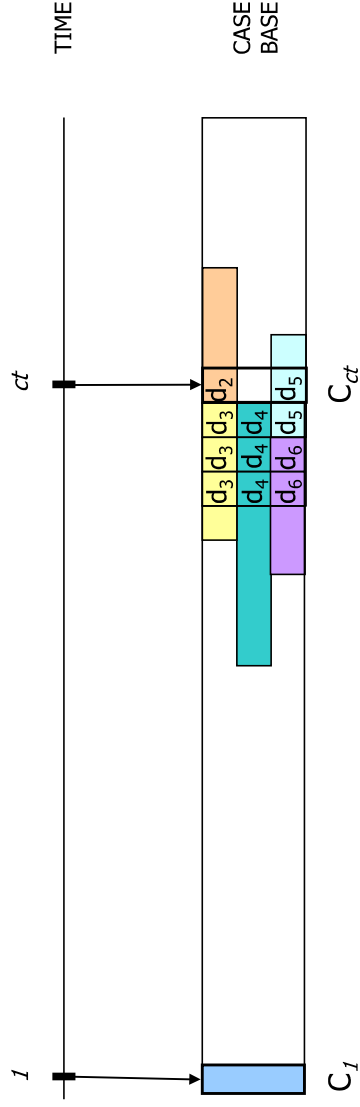
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Episode Retrieval (1)

[Sánchez-Marrè *et al.*, 2005]

- *Getting the possible diagnostics of the current case*
 - Get the closest metacases
- *Generation of possible episodes* arising from the current case
- For each *possible current episode*, most similar episodes must be retrieved.



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Episode Retrieval (2)

[Sánchez-Marrè *et al.*, 2005]

Input: C_{ct} , EpB , $MEpB$, CsB , $MCsB$

```

Sorted_Ep_L  $\leftarrow \emptyset$ 
 $CDL_{ct} \leftarrow \text{Get\_Diagnostics\_Current\_Case } (C_{ct}, MCsB)$ 
for each  $d \in CDL_{ct}$  do
  if  $(ct = 1)$  or  $(d \notin CDL_{ct-1})$  then    {new episodes}
    Retr_Ep  $\leftarrow \text{Retr\_Sim\_Episodes } (E_{ct,1}^d, MEpB, EpB)$ 
    Eval_Ep  $\leftarrow \text{Eval\_Sim\_Episodes } (E_{ct,1}^d, Retr\_Ep)$ 
    Sorted_Ep_L  $\leftarrow \text{Sorted\_Add } (Eval\_Ep, Sorted\_Ep\_L)$ 
  else                                     {continued episodes}
     $l \leftarrow \text{Comp\_Ep\_Length } (ct, d, CsB)$ 
    Retr_Ep  $\leftarrow \text{Retr\_Sim\_Episodes } (E_{ct-l,l+1}^d, MEpB, EpB)$ 
    Eval_Ep  $\leftarrow \text{Eval\_Sim\_Episodes } (E_{ct-l,l+1}^d, Retr\_Ep)$ 
    Sorted_Ep_L  $\leftarrow \text{Sorted\_Add } (Eval\_Ep, Sorted\_Ep\_L)$ 
  endif
endfor

```

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Episode Retrieval (3)

[Sánchez-Marrè *et al.*, 2005]

```

if  $ct \neq 1$  then
  for each  $(d \in \text{CDL}_{ct-1})$  and  $(d \notin \text{CDL}_{ct})$  do
     $l \leftarrow \text{Comp\_Ep\_Length}(ct, d, \text{CsB})$ 
     $\text{EpB} \leftarrow \text{EpB} + \text{Learn\_New\_Episode}(E_{ct-l}^d)$  {add a new Ep}
  endfor
endif

```

Returns: Sorted_Ep_L {First Ep is the most similar}

Episode Similarity (1)

[Sánchez-Marrè *et al.*, 2005]

Input: Ep_ct, Retr_Ep

```

Eval_Ep  $\leftarrow \emptyset$ 
for each  $\text{Ep} \in \text{Retr\_Ep}$  do
   $\text{Sim\_Degree\_Ep} \leftarrow \text{Episodic\_Sim}(\text{Ep}, \text{Ep\_ct})$ 
   $\text{Eval\_Ep} \leftarrow \text{Eval\_Ep} \cup \text{Build\_Pair}(\text{Sim\_Degree\_Ep}, \text{Ep})$ 
endfor

```

Returns: Eval_Ep

Episode Similarity (2)

[Sánchez-Marrè *et al.*, 2005]

$$Sim_{ep}(E_{t_1,l_1}^d, E_{t_2,l_2}^d) = \begin{cases} \frac{1}{l} \sum_{i=1}^l Sim_c(C_{t_1+i-1}, C_{t_2+i-1}) & \text{if } l_1 = l_2 = l \\ \frac{1}{\max(l_1, l_2)} \sum_{i=1}^{\min(l_1, l_2)} Sim_c(C_{t_1+i-1}, C_{t_2+i-1}) & \text{if } l_1 \neq l_2 \end{cases}$$

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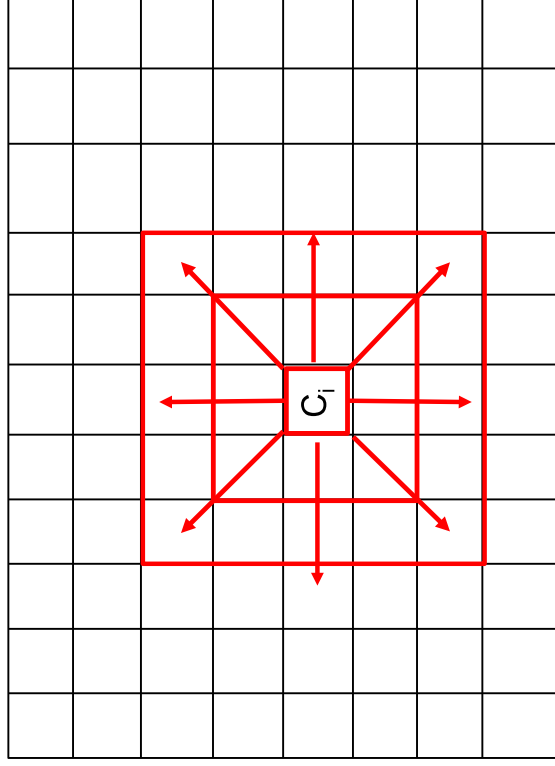
- Temporal CBR
- [Sánchez-Marrè *et al.*, 2005] M. Sànchez-Marrè, U. Cortés, M. Martínez, J. Comas and I. Rodríguez-Roda. An Approach for Temporal Case-Based Reasoning: Episode-Based Reasoning. *6th International Conference on Case-Based Reasoning (ICCBR'2005)*. ICCBR'2005 Proceedings. LNAI-3620, pp. 465-476. Chicago, USA. 2005.
- [Pascual-Pañach *et al.*, 2024] Josep Pascual-Pañach, Miquel Sànchez-Marrè, Miquel Àngel Cugueró-Escofet . A temporal case-based reasoning approach for performance improvement in intelligent environmental decision support systems. *Engineering Applications of Artificial Intelligence* Volume 136, Part A, October 2024
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Spatial CBR

Spatial CBR

- Problema:
 - tenim casos distribuïts en un espai
 - la relació espacial entre els casos pot ser rellevant durant el procés de Retrieve i/o d'Adapt
- Aproximacions:
 - Modificar la estructura de la llibreria de casos per incloure les relacions espacials
 - Introduir les relacions espacials com a part de la descripció del cas
 - Associar els casos amb un Sistema d'Informació Geogràfica (GIS)
 - De vegades anomenats Geo-CBR ó GIS-CBR

Aproximació 1: relacions espacials a la estructura de la llibreria de casos



Spatial Episodes: casos contigus a un cert cas, a una certa distància

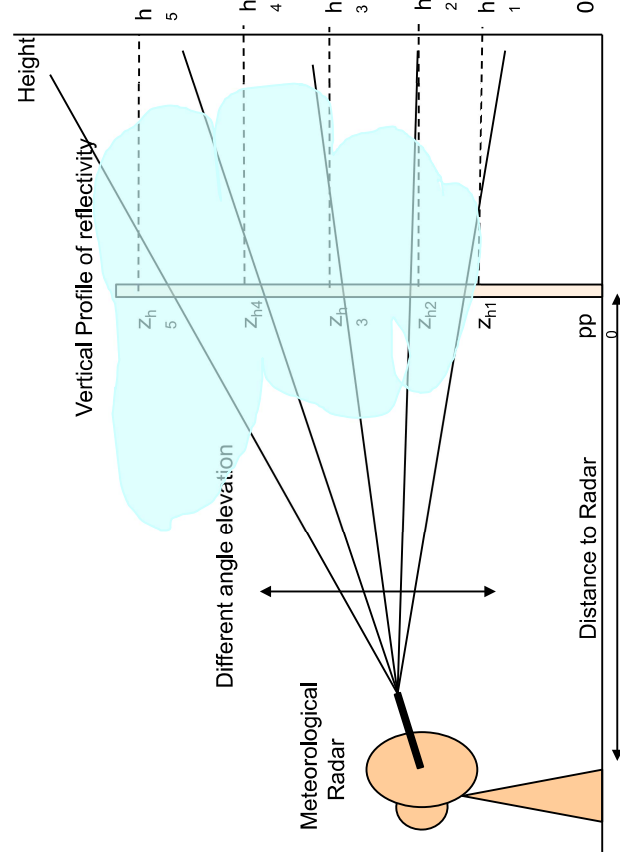
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Aproximació 1: relacions espacials a la estructura de la llibreria de casos

Exemple: Prevenció d'inundacions

[HYDRAMET Spanish Project REN2000-1755-C03-01]



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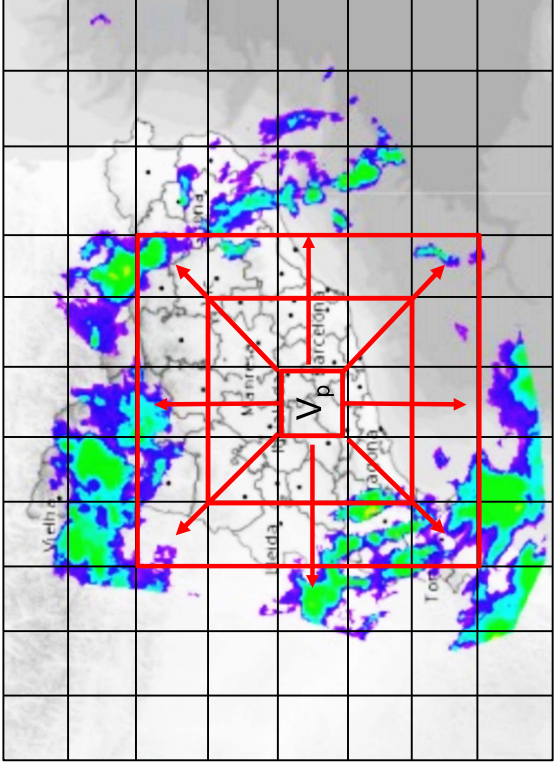
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Aproximació 1: relacions espacials a la estructura de

Exemple: Prevenció d'inundacions

[HYDRAMET Spanish Project REN2000-1755-C03-01]

Reflectivity (z): meteorological variable determined by the number and size of the particles presents in a sample volume



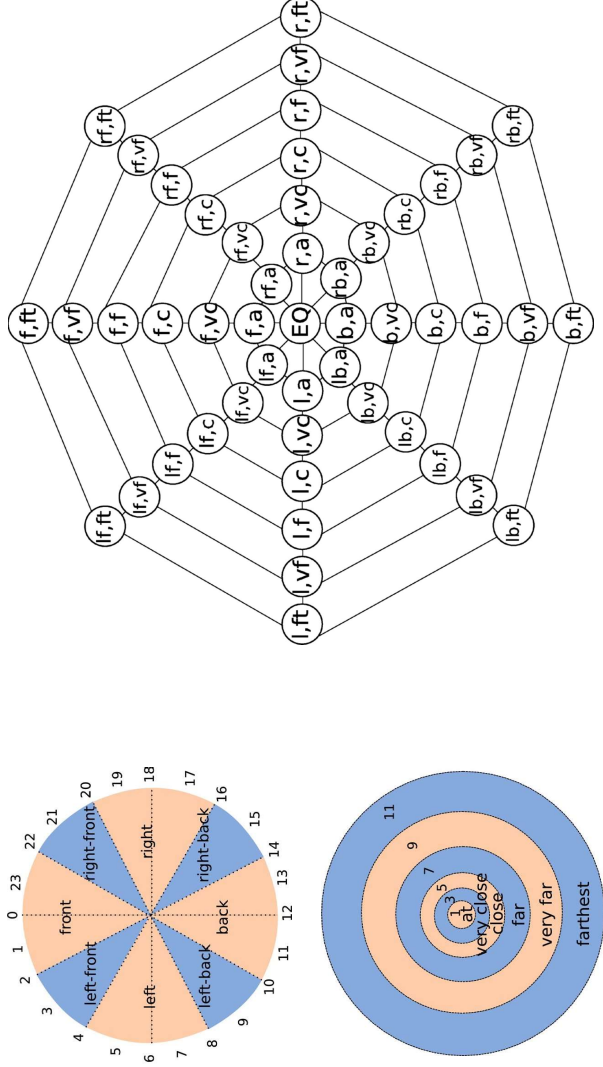
As the **rain-rate** increases:
the **reflectivity** increases:

Rainrate (in/h)	Reflectivity (dBz)
0.1	29.5
0.25	35.9
0.5	40.7
1.25	47.0
2.5	51.9
4.0	55.1

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Spatial CBR

Aproximació 2: Introduir les relacions espacials com a part de la descripció del cas



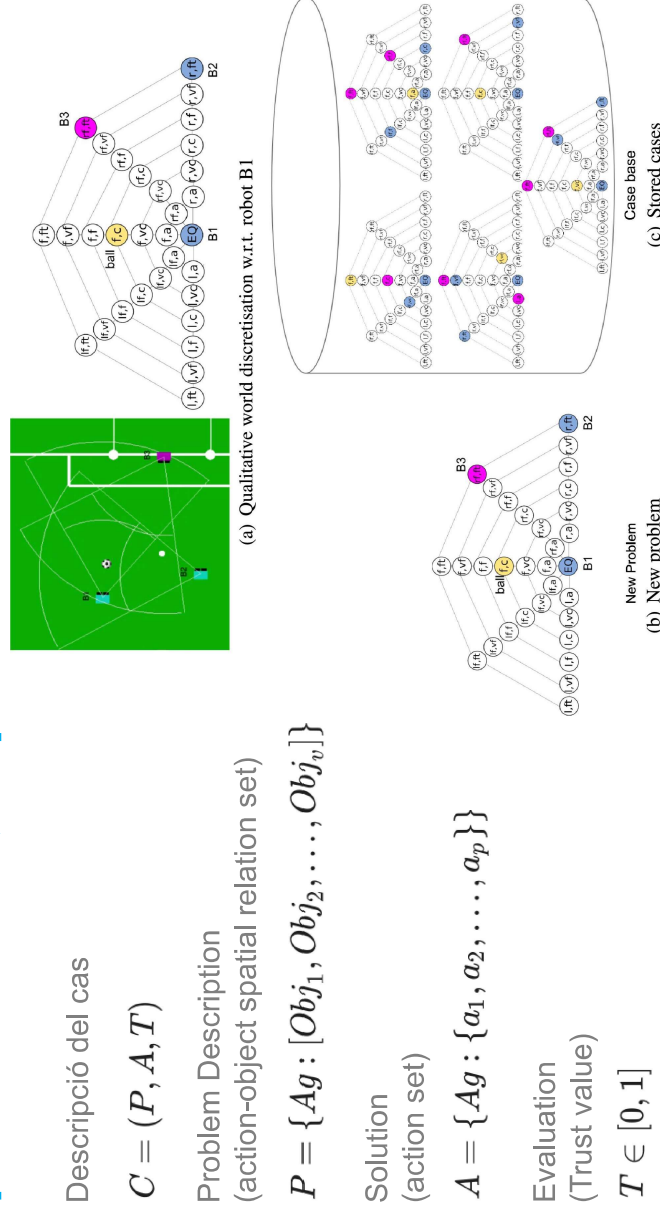
Relacions espais entre objectes basades en direcció (left, right, front, back...) i distància (very close, close, far, very far...)

images extracted from [Donadon Homem et al., 2020]

Aproximació 2: Introduir les relacions espacials com a part de la descripció del cas

Exemple: robo-soccer

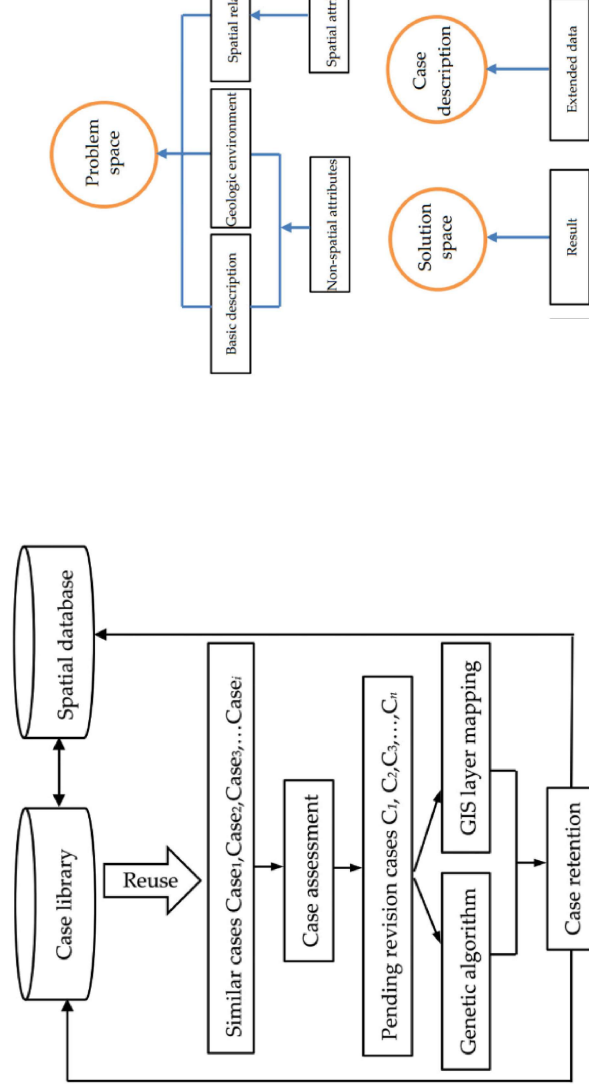
[Donadon Homem *et al.*, 2020]



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Aproximació 3: Associar els casos amb un Sistema d'Informació Geogràfica (GIS)

Geo-CBR | GIS-CBR



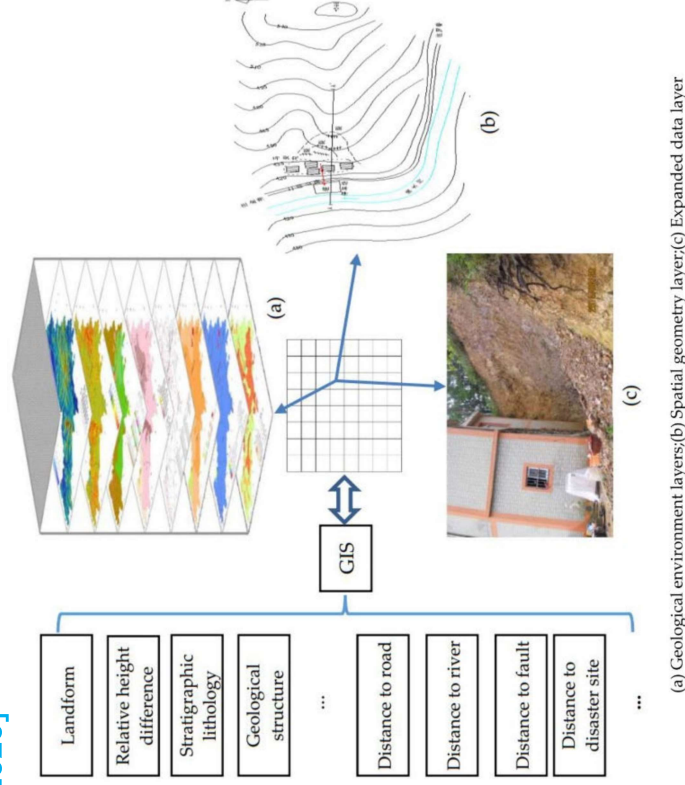
images extracted from [Deng & Li, 2020]

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Aproximacio 3: Associar els casos amb un Sistema d'Informació Geogràfica (GIS)

Exemple: Desastres Geològics

[Deng & Li., 2020]



(a) Geological environment layers;(b) Spatial geometry layer;(c) Expanded data layer

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Hybrid Neuro-Symbolic CBR

- IDEA: apply Artificial Neural Networks (ANNs) to one or several steps in the CBR Cycle

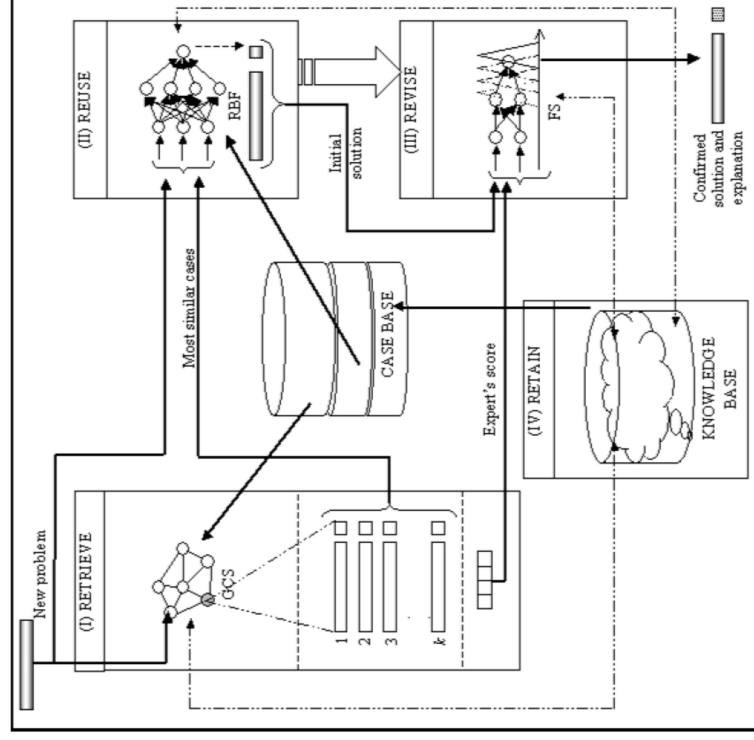
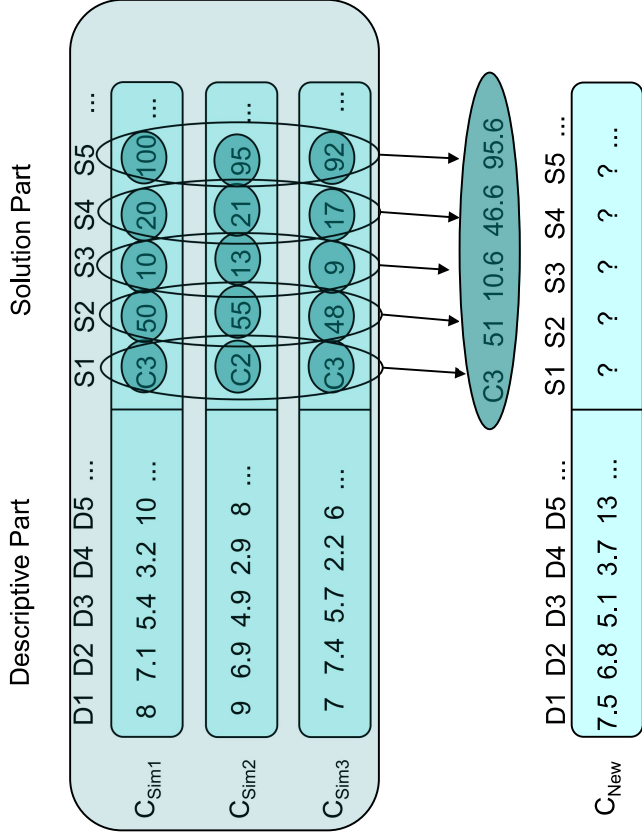


image extracted from [F. Fdez-Riverola & J.M. Corchado, 2003]

Hybrid Adaptation



- Each solution parameter can be predicted using different models (ANN, Multiple Linear Regression, Weighted mean, ...)

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An Example

Forecasting Red Tides Using an Hybrid Neuro-Symbolic System (1)
[F. Fdez-Riverola & J.M. Corchado, 2003]

- The aim is to develop a system for **forecasting one week in advance** the concentrations (in cells/liter) of the *pseudonitzschia spp*, the diatom that produces the most harmful **red tides**, at different geographical points.
- The forecasting problem can be stated as follows:
 - Given:** a sequence of data values (representative of the current and immediately previous state) relating to some physical and biological parameters,
 - Predict:** the value of a parameter at some future point(s) or time(s).

Table 1
Variables that define a case

Variable	Unit	Week
Date	dd-mm-yyyy	W_{n-1}, W_n
Temperature	Cent. degrees	W_{n-1}, W_n
Oxygen	milliliters/liter	W_{n-1}, W_n
PH	acidic/neutral	W_{n-1}, W_n
Transmittance	%	W_{n-1}, W_n
Fluorescence	%	W_{n-1}, W_n
Cloud index	%	W_{n-1}, W_n
Recount of diatoms	cel/liter	W_{n-1}, W_n
nitzschia spp	cel/liter	W_{n-1}, W_n
nitzschia spp (future)	cel/liter	W_{n+1}

An Example

Forecasting Red Tides Using an Hybrid Neuro-Symbolic System (2)

[F. Fdez-Riverola & J.M. Corchado, 2003]

CBR-STAGE	Technology	Input	Output	Process
Retrieval	GCS network.	Problem descriptor.	K cases.	All the cases that belong to the same class to which the GCS associates the Problem case.
Reuse	RBF network.	Problem descriptor.	Initial solution: concentration of pseudo-nitzschia spp.	The RBF network is retrained with the K retrieved cases.
Revision	4 Fuzzy systems.	Problem descriptor.	Confirmed solution: concentration of pseudo-nitzschia spp.	Four fuzzy systems are created using the RBF network configuration with different degrees of generalization.
Retain	GCS network, RBF network, 4 Fuzzy systems.	Problem descriptor, Forecasting Error.	Configuration parameters of the GCS network, RBF network and 4 Fuzzy systems.	The configurations of the GCS network, the RBF network and the Fuzzy subsystems are updated according to the accuracy of the forecast.

- GCS: Growing Cell Structures ANN
(a variation of Kohonen's Self-Organising Maps)
B. Fritzke. *Growing Self-Organizing Networks - Why?*. ESANN'96, Brussels, 61-72, 1996.
- RBF: Radial Basis Function ANN
B. Fritzke. *Fast learning with incremental RBF Networks*. Neural Processing Letters (1), num. 1, 2-5, 1994

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An Example

Forecasting Red Tides Using an Hybrid Neuro-Symbolic System (3)

[F. Fdez-Riverola & J.M. Corchado, 2003]

```

procedure retrieve (in  $v_x$ , GCS, out  $K$ )
{
01  $CD \leftarrow \emptyset$  /* vector of pairs(cell, distance) */
02 for each cell  $c \in GCS$ 
03     compute distance:  $d_c \leftarrow DIS(v_x, w_c)$ 
04     assign cell-distance pair:  $CD \leftarrow (c, d_c)$ 
05 order_by_distance(CD) /* ascending */
06 for each pair  $p \in CD$ 
07      $K \leftarrow get\_cases\_from\_cell(p)$ 
08     if  $|K| > 0$  then
09         go to line 10 /* non-empty cell */
10 end.
}
```

An Example

Forecasting Red Tides Using an Hybrid Neuro-Symbolic System (4)
[F. Fdez-Riverola & J.M. Corchado, 2003]

Table 6

Average error in the forecast with other techniques and the CBR-ANN-FS Hybrid System

Method	Type	Aver. error (cel/liter)
CBR-ANN-FS	Hybrid System	26.043,66
RBF	ANN	45.654,20
FIR	ANN	-
ARIMA	Statistics	71.918,15
Quadratic Trend	Statistics	70.354,35
Moving Average	Statistics	51.969,43
Simp. Exp. Smoothing	Statistics	41.943,26
Lin. Exp. Smoothing	Statistics	49.038,19

Table 7

Number of predictions with an error ≥ 100.000 cell/liter

Method	Incorrect predictions
CBR-ANN-FS	12
RBF	17
ARIMA	38
Quadratic Trend	14
Moving Average	20
Simple Exp. Smoothing	18
Brown's Linear Exp. Smoothing	25

Referències

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Explainable CBR (XCBR)

Explainable CBR (XCBR)

Tipus d'aplicacions de XCBR (1)

- Explainable CBR (XCBR) es divideix en dos tipus d'aplicacions:
 - Explicacions sobre altres sistemes d'IA basades en CBR
 - Explicacions sobre el procés del propi CBR

Tipus d'aplicacions de XCBR (2)

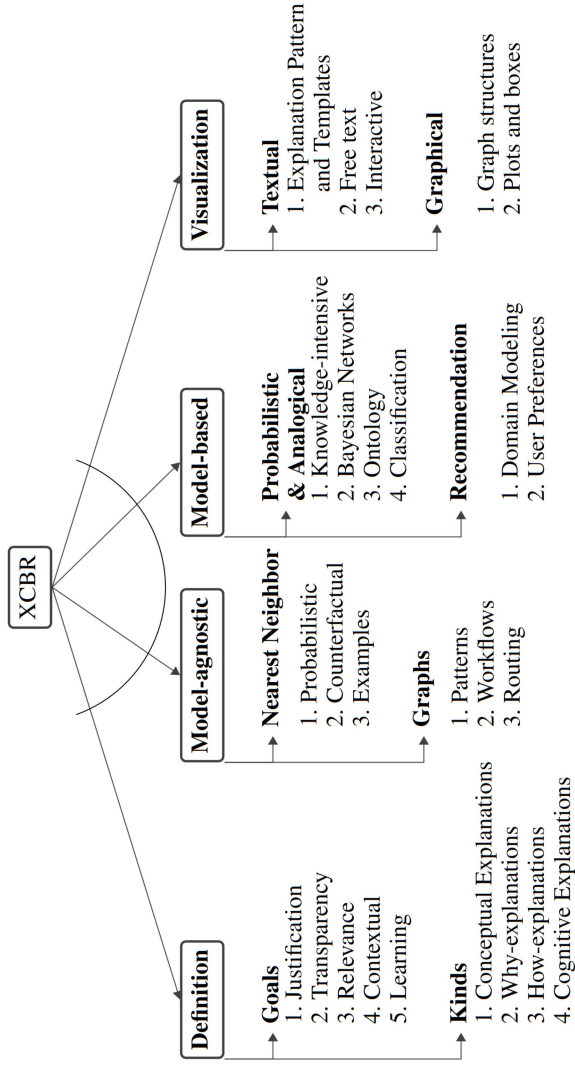
- Explicacions sobre altres sistemes d'IA
- Idea: L'ús de mètodes de raonament CBR com a tècnica d'explicació per obrir la caixa negra d'altres tècniques d'IA
 - com xarxes neuronals, SVM, algorismes genètics...
- Algunes aproximacions:
 - Ús de l'experiència i tècniques basades en la memòria per generar explicacions.
 - Recuperar i reutilitzar explicacions humanes.
 - Ús de sistemes bessons on el mòdul CBR dupliqui o col·labora amb el model d'IA de caixa negra.
 - CBR com a sistema de raonament de segona opinió en un procés de raonament diferent.

Tipus d'aplicacions de XCBR (3)

- Explicacions sobre el procés CBR
- Idea: explicar processos complexos dins el raonament del sistema CBR
- Algunes aproximacions
 - Dotar d'explicabilitat als passos del CBR
 - Explicar mesures de semblança
 - Explicar processos d'adaptació.
 - Explicar processos d'oblit
 - Explicacions sobre el coneixement del domini
 - Explicacions intenses en coneixement basades en recursos externs.

Taxonomia d'explicacions en sistemes XCBR

[Schoenborn *et al.*, 2021]



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